LA HW

Included exercises: 1, 3, 7, 9, 13, 17, 21, 23, 25, 27, 29, 31, 39, 43, 49, 51, 53, 59, 61, 63, 69, 71, 75, 79

Exercises 1–12: Dependence by Inspection

Determine by inspection whether each given set is linearly dependent.

Original Exercise 1

$$\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\}.$$

Original Exercise 3

$$\left\{ \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 6 \\ 0 \end{bmatrix} \right\}.$$

Original Exercise 7

$$\left\{ \begin{bmatrix} -3 \\ 12 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \end{bmatrix} \right\}.$$

Original Exercise 9

$$\left\{ \begin{bmatrix} 5 \\ 3 \end{bmatrix} \right\}.$$

Exercises 13–22: Smallest Subset with the Same Span

For each set S, determine by inspection a subset containing as few vectors as possible and having the same span as S.

Original Exercise 13

$$S = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ -6 \end{bmatrix} \right\}.$$

Original Exercise 17

$$S = \left\{ \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \\ 10 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

Original Exercise 21

$$S = \left\{ \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix} \right\}.$$

Exercises 23–30: Linear Independence

Determine whether each given set is linearly independent.

Original Exercise 23

$$\left\{\begin{bmatrix}1\\-1\\-2\end{bmatrix},\begin{bmatrix}-1\\0\\1\end{bmatrix},\begin{bmatrix}1\\2\\1\end{bmatrix}\right\}.$$

Original Exercise 25

$$\left\{\begin{bmatrix}1\\2\\0\\-1\end{bmatrix},\begin{bmatrix}1\\-3\\1\\-2\end{bmatrix},\begin{bmatrix}1\\2\\-2\\3\end{bmatrix}\right\}.$$

Original Exercise 27

$$\left\{\begin{bmatrix}1\\0\\0\\-2\end{bmatrix},\begin{bmatrix}0\\1\\-1\\0\end{bmatrix},\begin{bmatrix}1\\0\\1\\1\end{bmatrix},\begin{bmatrix}0\\1\\0\\1\end{bmatrix}\right\}.$$

Original Exercise 29

$$\left\{\begin{bmatrix}1\\-1\\-1\\2\end{bmatrix},\begin{bmatrix}-1\\0\\1\\-1\end{bmatrix},\begin{bmatrix}-1\\-4\\1\\3\end{bmatrix},\begin{bmatrix}0\\1\\-2\\1\end{bmatrix}\right\}.$$

Exercises 31–38: Expressing One Vector Using the Others

Each set S is linearly dependent. Write some vector in S as a linear combination of the others.

Original Exercise 31

$$S=\left\{\begin{bmatrix}-1\\1\\2\end{bmatrix},\begin{bmatrix}3\\-3\\-6\end{bmatrix},\begin{bmatrix}0\\1\\2\end{bmatrix}\right\}.$$

Exercises 39–50: Parameterized Dependence

Determine, if possible, a value of r for which the given set is linearly dependent.

Original Exercise 39

$$\left\{\begin{bmatrix}1\\-1\end{bmatrix},\begin{bmatrix}-3\\3\end{bmatrix},\begin{bmatrix}4\\r\end{bmatrix}\right\}.$$

Original Exercise 43

$$\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \end{bmatrix}, \begin{bmatrix} r \\ 0 \end{bmatrix} \right\}.$$

Original Exercise 49

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 6 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -2 \\ r \end{bmatrix} \right\}.$$

Exercises 51–62: Homogeneous Systems

Write the vector form of the general solution of each system.

Original Exercise 51

$$x_1 - 4x_2 + 2x_3 = 0.$$

Original Exercise 53

$$\begin{aligned} x_1 + 3x_2 + 2x_4 &= 0, \\ x_3 - 6x_4 &= 0. \end{aligned}$$

Original Exercise 59

$$\begin{aligned} x_1 + x_2 + x_4 &= 0, \\ x_1 + 2x_2 + 4x_4 &= 0, \\ 2x_1 - 4x_4 &= 0. \end{aligned}$$

Original Exercise 61

$$\begin{aligned} x_1 + 2x_2 - x_3 + 2x_5 - x_6 &= 0, \\ 2x_1 + 4x_2 - 2x_3 - x_4 - 5x_6 &= 0, \\ -x_1 - 2x_2 + x_3 + x_4 + 2x_5 + 4x_6 &= 0, \\ x_4 + 4x_5 + 3x_6 &= 0. \end{aligned}$$

Exercises 63–82: True or False

Determine whether each statement is true or false.

Original Exercise 63 If S is linearly independent, then no vector in S is a linear combination of the others.

Original Exercise 69 A homogeneous equation always has infinitely many solutions.

Original Exercise 71 For any vector $\mathbf{v}$, the set $\{\mathbf{v}\}$ is linearly dependent.

Original Exercise 75 For the system $A\mathbf{x} = \mathbf{b}$ to be homogeneous, $\mathbf{b}$ must equal $\mathbf{0}$.

Original Exercise 79 If $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k = \mathbf{0}$ for $c_1 = c_2 = \cdots = c_k = 0$, then $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is linearly independent.